

Rubric Assessment Summary for CHE 396 and CHE 397

Rubric-based assessment was conducted for two selected semesters of CHE 396 and CHE 397 to evaluate SLO 2 and SLO 5. All assessments were based on team submissions (groups of 4–5 students).

For CHE 396, assessment was conducted using Mini-Project 3, which focuses on process economics (process safety is assessed separately in another assignment). The average score for SLO 2 was 2.6, indicating performance between improving and satisfactory. Students demonstrated strong economic analysis skills; however, broader considerations such as environmental and societal impacts were not emphasized in this assignment. For SLO 5, the average score was 2.6, reflecting generally effective teamwork, though some variability in participation and leadership was observed.

For CHE 397, assessment was based on a final project presentation, which integrates process design, economics, safety, and control. The average score for SLO 2 was 2.7, indicating satisfactory performance with improved integration of multiple design constraints, including safety and economic considerations. The average score for SLO 5 was 2.7, showing strong team collaboration, organization, and goal achievement across most groups.

Key Findings

- Students demonstrate strong technical competency
- Integration of broader design considerations improves in CHE 397
- Teamwork is generally effective, with minor imbalances in contribution

Recommendations

- Increase emphasis on environmental and societal considerations in earlier design work (CHE 396)
- Continue reinforcing safety and control integration in CHE 397
- Support team effectiveness through structured peer evaluation and role clarity

Overall, student performance meets expectations, with clear progression from CHE 396 to CHE 397 in both design integration and teamwork.